Questions by topic: Biogeochemical cycles: carbon, nitrogen, phosphorus - Set 1 (75 marks)

Answer all questions in the spaces provided.

0 1 Table 1 shows the mass of carbon in the main reservoirs of the carbon cycle.

Table 1

Reservoir	Mass of carbon / billion tonnes
Atmosphere (CO_2)	700
Living organisms	470
Dead organic matter (soil)	3 700
Surface ocean waters	500
Deep ocean (dissolved)	35 000
Fossil fuels	10 000
Sedimentary carbonate rocks	20 000 000

0 1 . 1 Express the mass of carbon in the atmosphere in standard form, in tonnes.

[1 mark]

Answer _____ tonnes

0 1 . 2 Calculate the percentage increase in atmospheric carbon that would result from burning 10% of the fossil fuel reservoir, assuming all the carbon remained in the atmosphere.

[2 marks]

Answer _____ %

0 1 . 3 Explain why the sedimentary carbonate rocks, although by far the largest reservoir, have little effect on atmospheric CO_2 concentrations.

[2 marks]

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0 1 . 4 Describe how carbon moves from the atmosphere into sedimentary carbonate rocks.

[3 marks]

0 1 . 5 Explain why aerobic respiration returns carbon to the atmosphere as CO_2 but anaerobic respiration returns it mainly as methane.

[2 marks]

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0 2 . 1 Explain how each of the following human activities changes the amount of carbon in the atmosphere.

[6 marks]

Ploughing farmland .....

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Draining marshland / peat bogs .....

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Toxic pollution of the sea .....

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0 2 . 2 Give **three** anaerobic environments created by human activities that release methane.

[3 marks]

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3 .....

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0 2 . 3 Explain how rising atmospheric CO_2 causes ocean acidification.

[1 mark]

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0 3 . 1 Explain why planting a new forest only sequesters carbon for a limited period, and how the storage can be extended.

[3 marks]

0 3 . 2 Compare pre-combustion and post-combustion methods of carbon capture.

[4 marks]

0 3 . 3 Describe **two** ways captured CO_2 can be stored and give **one** concern about long-term storage.

[3 marks]

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0 4 Figure 1 shows a simplified nitrogen cycle with some processes labelled 1 to 6.

Figure 1

Number	Process description
1	Nitrogen gas converted to ammonium ions by Rhizobium in root nodules
2	Ammonium ions oxidised to nitrite then nitrate by Nitrosomonas and Nitrobacter
3	Nitrates reduced to nitrogen gas by Pseudomonas in anaerobic soil
4	Amino groups in dead organic matter released as ammonium ions by decomposers
5	Nitrogen and oxygen react during lightning to form NOx
6	Nitrates washed out of soil into rivers

0 4 . 1 Name processes 1 to 6.

[3 marks]

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0 4 . 2 Explain why draining a waterlogged field increases the nitrate available to crops.

[3 marks]

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0 4 . 3 Explain why nitrogen is described as a resource that is 'not very abundant' for living organisms even though the atmosphere is 78% nitrogen.

[2 marks]

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0 4 . 4 Explain why growing legumes such as clover in a crop rotation reduces the need for artificial fertiliser.

[2 marks]

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0 5 Table 1 shows global reactive nitrogen fixed each year by natural processes and by human activities.

Table 1

Source	Nitrogen fixed / million tonnes per year
Biological fixation (natural ecosystems)	110
Lightning	5
Haber process (fertiliser manufacture)	120
Cultivation of legume crops	60
Combustion (NOx)	30

0 5 . 1 Use Table 1 to calculate the percentage of total nitrogen fixation that is caused by human activities.

[2 marks]

Answer ..... %

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0 5 . 2 Describe the Haber process and explain why it is unsustainable compared with biological nitrogen fixation.

[3 marks]

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0 5 . 3 Describe **three** consequences of the increased movement of nitrogen compounds caused by human activities.

[3 marks]

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0 5 . 4 Explain how composting or anaerobic digestion of biological wastes produces a better fertiliser than raw waste.

[2 marks]

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0 6 . 1 Explain why phosphorus is more likely than nitrogen to be a limiting factor on biological productivity.

[3 marks]

0 6 . 2 Explain how mycorrhizal fungi increase phosphate uptake by plants.

[2 marks]

0 6 . 3 Describe how human activities have altered the phosphorus cycle.

[3 marks]

0 6 . 4 Explain **two** strategies for the sustainable management of phosphorus supplies.

[2 marks]

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0 7 . 1 Describe how human activities have altered the natural equilibria of the carbon and nitrogen cycles, and evaluate the strategies that could counteract these changes.

[9 marks]

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0 7 . 2 State what is meant by the **residence time** of a substance in a reservoir.

[1 mark]

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0 8 A student measured the mass of carbon stored per hectare in three land uses in the same region.

Table 1

Land use	Carbon in vegetation / t ha^{-1}	Carbon in soil dead organic matter / t ha^{-1}
Mature woodland	120	95
Permanent pasture	3	80
Arable (ploughed annually)	2	45

0 8 . 1 Calculate the total carbon lost per hectare when mature woodland is converted to arable farmland.

[1 mark]

Answer ____ t/ha

0 8 . 2 Explain why the soil carbon in arable land is so much lower than in pasture, even though the vegetation carbon is similar.

[2 marks]

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0 8 . 3 Suggest **two** ways the arable farmer could increase soil carbon storage.

[2 marks]

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2 .....

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END OF QUESTIONS

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